

Hello. I'm looking to have a Web app created where users can go to the website, sign on with their member account, and then use the web app within the site to assist them with US Government contract solicitations.

All federal contract opportunities for the US government are listed at www.sam.gov and typically anyone who is registered with the government to do business with the government can go onto sam.gov to find which contract opportunities make the most sense for the business. Once you click on one of the contract opportunities, typically there will be a section on the page that list any document attachments. Those document attachments typically are the solicitation documents.

Project Overview

I need a web application that automatically analyzes government contracting solicitation documents and extracts critical procurement information to streamline the bid preparation process.

Application Startup Workflow (CRITICAL)

Initial User Interface

When the application is launched, it MUST immediately present the user with:

1. Primary Input Field: SAM.gov solicitation URL (required field)
2. Secondary Optional Upload: "Upload additional PDF documents related to this solicitation" (optional)
3. Begin Analysis Button: Once URL is entered (and optional documents uploaded), user clicks to initiate analysis and automations

Flow: User provides SAM.gov link → (optionally uploads PDFs) → clicks "Analyze" → application begins automated extraction, research, and processing

Core Functionality Required

Document Input Methods

The application must support TWO input methods:

1. SAM.gov Integration (PRIMARY/REQUIRED): Accept SAM.gov opportunity URL, analyze the page, and automatically download all attached documents from the solicitation
2. Optional Supplemental Upload: Accept additional solicitation documents (PDF/Word) uploaded by user

Solicitation Type Classification (REQUIRED)

The application must automatically analyze and classify each SAM.gov opportunity as:

- Product-based solicitation
- Service-based solicitation
- Both product and service-based solicitation

Display this classification prominently in the analysis results.

1. Deadline Extraction

- Capture submission due date and time
- Include timezone if specified in document

2. Product/Service Details (CLIN Analysis)

For each Contract Line Item Number (CLIN), extract:

- Product name and description
- Manufacturer name
- Part/model number
- Quantity required
- For services: scope of work, timeline, and service requirements

3. Delivery Requirements

- Complete delivery address
- Special delivery instructions
- Required delivery date

4. Manufacturer Research (Automated)

- Find manufacturer's official website
- Locate and extract sales team contact email

5. Dealer/Distributor Research (Automated)

Identify top 8 authorized dealers/distributors/service providers:

- Company name
- Website URL
- Sales contact email
- Current retail pricing (if publicly available)

6. Email Template Generation with Review Capability

Auto-generate professional quote inquiry emails with:

- Product specifications (name, manufacturer, part number, quantity)
- Delivery address for shipping calculations
- Required delivery deadline
- Request for product datasheets/spec sheets
- Request for Net payment terms

Email Tone Requirements:

- Opening: "Hello, We're currently working on a project and would like to request a quote"
- Strategic language that conveys we're evaluating multiple competitive quotes
- Professional but relaxed and assertive tone
- CRITICAL: Never reference government contracts or solicitations
- Maintain standard commercial inquiry appearance

Email Review Process (REQUIRED):

- BEFORE any emails are sent, the application MUST present ALL generated emails to the user for review
- User interface should display each email with:
- Recipient name and email address
- Subject line

- Full email body
- Edit capability for each email
- User must have ability to:
- Approve individual emails
- Edit any email before sending
- Select which emails to send (checkboxes)
- Send all approved emails in batch
- NO emails should be sent without explicit user approval

Critical Integration Requirements

Email Integration (REQUIRED)

The application MUST integrate with:

- Gmail (full OAuth integration)
- Microsoft Outlook (full OAuth integration)

Purpose: Automate sending of quote inquiry emails directly to identified dealers/distributors from user's email account AFTER user review and approval.

Calendar Integration (REQUIRED)

The application MUST integrate with:

- Google Calendar
- Apple iCal
- iCloud Calendar
- iPhone Calendar
- Microsoft Outlook Calendar

Purpose: Automatically create calendar events for extracted solicitation due dates/deadlines from analyzed SAM.gov opportunities and uploaded documents.

PDF/Document Editing Integration (REQUIRED)

The application MUST include integrated document editing capability:

- Adobe Acrobat integration OR equivalent high-capability PDF editor
- Word document editing capability
- Support editing of government forms (SF1449, etc.)

Key Features Required:

- Autofill Form Feature: Automatically populate government contract forms (SF1449, etc.) with extracted data and saved user/company information
- In-app editing of PDF and Word documents without downloading
- Save edited documents within the application
- Support for fillable PDF forms

Purpose: Allow users to complete bid submissions directly within the web app using extracted solicitation data to auto-populate required forms.

Output Format

Present all findings in clearly organized sections with headers. When information cannot be found, clearly indicate “Not found in provided documents” or “Requires additional research.”

Technical Requirements

- Streamlined startup interface with SAM.gov URL as primary input
- Accept optional supplemental PDF and Word document uploads
- SAM.gov URL input with automated attachment downloading
- AI/ML capability to classify solicitation types (product/service/both)
- Automated web scraping/research capability for manufacturer and dealer information
- Gmail and MS Outlook OAuth integration for automated email sending
- Email review and approval interface before sending
- Multi-calendar platform integration (Google, Apple, iCloud, Outlook)
- Integrated PDF/Word editor with autofill capabilities for government forms

- Clean, user-friendly interface with clear workflow progression
- Export capability for extracted data and email templates
- Secure handling of authentication credentials
- Document storage and management system

Use Case

This tool will be used for analyzing federal government solicitations to quickly prepare competitive bids while maintaining efficient vendor communications that secure best pricing without revealing government contract context. The automation of email outreach, deadline tracking, and bid form completion will significantly streamline the entire bid preparation workflow from analysis through submission, while maintaining user control over all outgoing communications.

What I Need From You

Please provide:

1. Estimated timeline for development
2. Your cost estimate
3. Technology stack you'll use (especially for SAM.gov scraping, calendar/email integrations, and PDF editing)
4. Experience with OAuth integrations, government website scraping, and PDF form manipulation
5. Approach to autofill functionality for government forms
6. User interface mockups or wireframes (if possible)
7. Any clarifying questions about functionality